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Post-Fall Decision Tree Development and Implementation

Bonita M. Gordon, MSN, RN, ANP-BC, ACNS-BC, CNRN; Theresa Frissora Wnek, MSN, RN; Nancy Glorius, MEd, RN; Carmen Hasdorff, MSN, RN; Joyce Shiverski, BSMNS, RN, NE-BC; Janet Ginn, BSMNS, RN, NE-BC

Care and evaluation after a patient's fall require a number of steps to ensure that appropriate care is given and injury is minimized. Astute and appropriate assessment skills with strategic interventions and communication can minimize the harm from a fall. Post-Fall Decision Guidelines were developed to guide care and treatment and to identify potential complications after a patient has fallen. This systematic approach mobilizes the steps of communication, using the Situation-Background-Assessment-Recommendation (SBAR) format, and guides assessment interventions. **Key words:** *assessment*, *communication*, *fall*, *fall prevention*, *guidelines*, *SBAR*

IN people 65 years and older, falls are the leading cause of injury or death.[1] Among older persons, falls are the underlying cause of a large proportion of fatal traumatic brain injuries that caused nearly 8000 deaths and 56 000 hospitalizations in 2005.[2] Falls account for a significant portion of injuries in hospitalized patients. Various precautions and interventions should be employed to prevent falls; equally important are the assessments and interventions that occur after a patient has fallen. Astute assessment skills, timely interventions, and thorough communication work to minimize harm after a fall has occurred. While exploring and broadening understanding of appropriate care after a fall, a clearly articulated method to guide the communications, physiological assessments, and interventions necessary to complete the care of a patient who experienced a fall could not be identified. Our literature search produced 43 articles with only 1 article addressing postfall care of the patient to reduce the risk of further injury. This one article was of an institution's implementation of an evidence-based practice. The literature review demonstrated an absence of formalized postfall assessments.

Post-Fall Decision Guidelines were created as the means to address the actions that are to be taken in a systematic manner after a patient has fallen. These guidelines are a reminder for the direct care provider to consider a patient's underlying condition by clarifying what patient information to seek out and share. They also offer a framework for a targeted assessment based on the area that is potentially injured. The relationship between the patient's history, injuries, and targeted assessment is captured with the 1-page Post-Fall Decision Guidelines.

WHY DEVELOP A TOOL

A review of findings from The Joint Commission indicates that patient falls are the sixth most commonly reported sentinel event, in which the patient suffers permanent harm or loss of function or dies of the injury sustained in the fall.[3] In the time period of January 1995 to September 30, 2009, The Joint Commission recorded 406 sentinel events involving patient falls accounting for

Author Affiliation: *Bethesda North Hospital—TriHealth, Cincinnati, Ohio.*

Corresponding Author: *Bonita M. Gordon, MSN, RN, ANP-BC, ACNS-BC, CNRN, Bethesda North Hospital—TriHealth, 10500 Montgomery Rd, Cincinnati, OH 45245 (bonita_gordon@trihealth.com).*

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6.3% of all reported sentinel events.[3] Recognizing the potential serious outcomes that can follow a patient fall, The Joint Commission developed National Patient Safety Goal 9 to reduce the risk of patient harm resulting from falls. As part of this goal, hospitals need to develop a fall-reduction program that includes the following elements: assess the fall risk of the patient, develop interventions based on the fall-risk assessment, train all staff members in the appropriate interventions to prevent falls, and monitor the effectiveness of the program.

Falls pose an even greater threat to patient safety when coupled with other high-risk conditions such as the use of anticoagulation medications, coagulation disorders, and osteoporosis. The increased risk of hemorrhage from the fall injury in patients using anticoagulation medications requires skilled and frequent assessment of the patient following a fall and decisive action with any early signs of deterioration.

The Post-Fall Decision Guidelines provide staff a standardized method of assessment, communication, and intervention following a patient fall. The guidelines are a checklist and serve as a decision aid for both nursing staff and physicians. Applying a readily available model through a visual system minimizes relying on memory alone and can refine decision making.[4] The guidelines also facilitate overseeing the multiple tasks required and prevent decision-making errors and omissions in care.[5] Autonomy among practitioners can lead to variations in practice. The guidelines foster a common understanding of expectations in care. Medical processes are often customized as pertaining to each practitioner; a key is to simplify systems and processes so that they can be performed consistently.[6]

COMPONENTS OF THE POST-FALL DECISION GUIDELINES

A quality improvement team of staff nurses, managers, a clinical nurse specialist, a quality improvement reviewer, and educators was formed. The team was given the task of reviewing all elements of a patient's fall and identifying steps that should occur after a fall. The steps were not obvious and the processes proved challenging. It became evident that multiple approaches in assessment were called for on the basis of the anatomical area injured. Although attention should focus on fall prevention, not every fall can be prevented. The literature review assisted the team in developing a decision tree to guide the care of a patient after a fall.

The Post-Fall Decision Guidelines start with an initial assessment of the patient who sustained a fall to determine the potential severity of any injury. The expectation is for the nurse to assess the patient at the point of the fall and examine any injured areas. The nursing assessment includes vital signs, with a comparison with baseline, and assessment of mental status, degree of pain, and any obvious injury or deformity. The guidelines also include assessment of clotting function. This is relevant because a fall in a patient using anticoagulation medications or a patient with low platelets carries increased risk of morbidity and mortality.[7] Included in the guidelines are specific considerations for the prospective traumatic brain injuries or hip fracture because of their potential morbidity and mortality (Figure 1). The nurse is then guided to immediately notify the charge nurse, who is the first level in the chain of command, and can ensure additional assistance to help the patient back to bed.

Communication of individual patient factors between teams of caregivers is crucial. Sentinel event data from The Joint Commission estimated that communication failure was the root cause in 70% of 2400 safety cases that resulted in patient harm.[3] To ensure complete communication, information is relayed to the physician, using the Situation-Background-Assessment-Recommendations (SBAR) format, which is a simple framework for organizing and delivering vital communication about a patient's condition.[8] This communication method is recognized by The Joint Commission as a way to meet the

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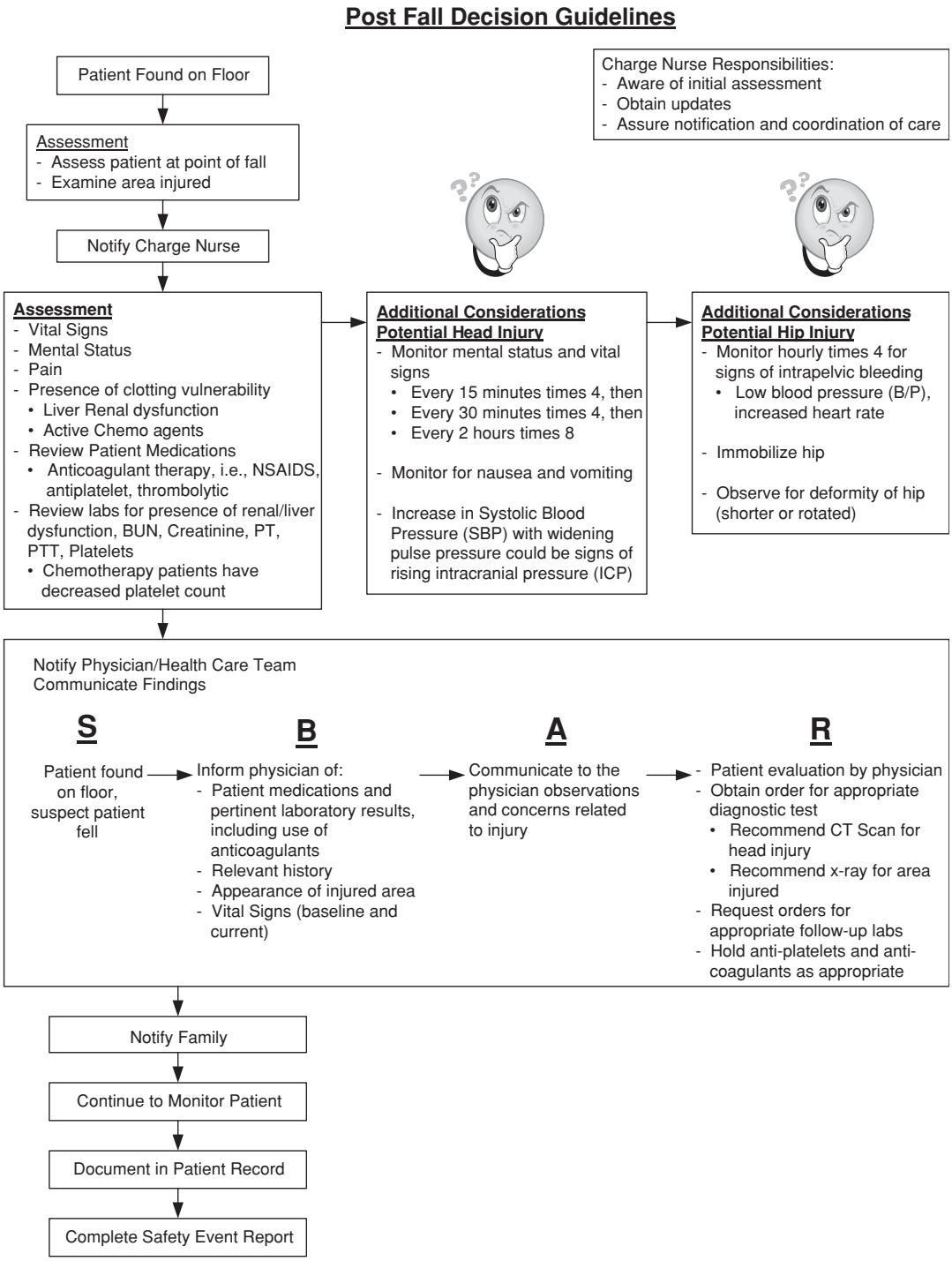


Figure 1. Post-Fall Decision Guidelines. NSAIDs indicates nonsteroidal anti-inflammatory drugs; BUN, serum urea nitrogen; PT, prothrombin time; and PTT, partial thromboplastin time. Reprinted with permission. Copyright 2009 TriHealth, Inc.

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standards included in the National Patient Safety Goal, which requires hospitals to improve communication between caregivers.[9] The communication regarding a patient fall would follow the following format:

S—Situation, for example, the patient fell from the bed to the floor

B—Background including pertinent information, for example, admitting diagnosis, medications, laboratory values

A—Assessment of injury, immediate vital signs, and other data

R—Recommend interventions

Using the SBAR format, the nurse would report the Situation of a patient fall with possible injury. The nurse would communicate pertinent Background information including history of liver or renal disease, coagulation disorders and whether disease or drug related, and present vital signs and mental status. The nurse would then present the Assessment of potential injuries, particularly head or hip related, and make Recommendations for interventions such as head computed tomographic scan or hip radiography, holding anticoagulation medication, obtaining laboratory tests, and giving platelets. When the physician examines and assesses the patient, these points are discussed. The SBAR format provides an organized and focused communication method and is in the guidelines to assist the nurse with this communication after a fall.[10]

The nurse also communicates to the patient's family or next of kin that a fall has occurred and relates precautions being taken. The nurse's actions, assessments, and communications are documented, and the nurse completes a Safety Event Report of the incident.

HOW POST-FALL DECISION GUIDELINES WERE IMPLEMENTED

The Post-Fall Decision Guidelines were introduced at a bimonthly meeting of nurse leaders. The guidelines were first presented at the July 2008 meeting, and copies of the Post-Fall Decision Guidelines were distributed to nurse managers and other nursing leaders. The guidelines were also posted on the hospital intranet, which is available to all staff members. Poe et al[11] noted that staff members often did not access their "fall prevention" documents electronically; therefore, laminated copies were provided to post on the nursing units.

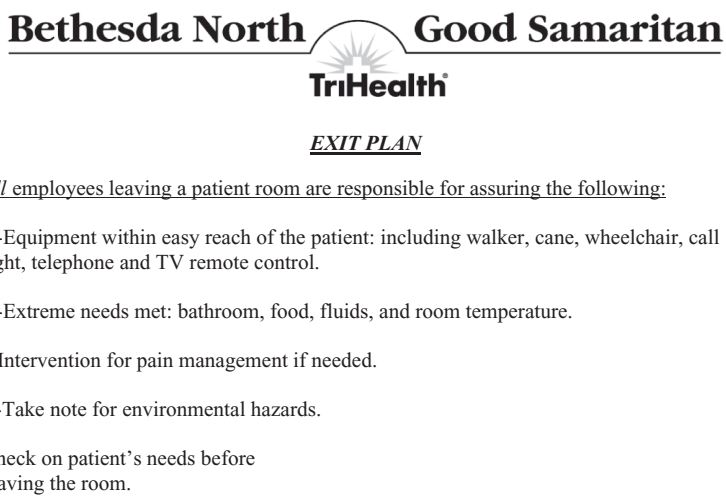
The information was presented to the direct caregivers as part of 2 quarterly mandatory education packets. The information provided in the education packets was also available on the hospital intranet in the eLearning system. As proposed by Kramer and Schmalenberg,[12] staff satisfaction is strongly related to their ability to provide good quality care and the ability to depend on coworkers to provide that same quality of care. This requires organizations to provide easy access to education and training. This was accomplished through our internal eLearning program.

Initially, the education about the Post-Fall Decision Guidelines included an explanation about the impact of patient falls and importance of preventing them, information from recent research studies on complications from patient falls, case studies of patient falls, examples of documentation of falls based on these case studies, and the guidelines themselves. A postquiz was required. The second education segment on the Post-Fall Decision Guidelines focused on fall prevention documentation and the use of new screens developed in our electronic clinical record to facilitate this documentation. The documentation screens closely followed the format of the Post-Fall Decision Guidelines, reinforcing adherence to the guidelines. A reference to the Post-Fall Decision Guidelines and how to access them on the hospital intranet was included in this education.

OUTCOMES

Deming's performance improvement model Plan-Do-Check-Act (PDCA) was used throughout the development of the

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EXIT PLAN

All employees leaving a patient room are responsible for assuring the following:

E-Equipment within easy reach of the patient: including walker, cane, wheelchair, call light, telephone and TV remote control.

X-Extreme needs met: bathroom, food, fluids, and room temperature.

I-Intervention for pain management if needed.

T-Take note for environmental hazards.

Check on patient's needs before leaving the room.

Figure 2. Exit plan to reduce the risk of falls. Reprinted with permission. Copyright 2009 TriHealth, Inc.

guidelines.[13] A pilot of the guidelines was trialed on 1 unit, which resulted in adjustments to the wording. A medical record review of patients who had fallen was performed using a gap analysis tool. Of the 33 medical records reviewed, all the Post-Fall Decision Guidelines were implemented for all patients and documentation on the electronic screen for postfall. Of note, additional information was included in the documentation such as sent for stat head computed tomography, no head or hip injury, or on heparin. This review confirmed that the actions and assessments performed by the nurse were consistent with what was recommended in the guidelines. Test scores from the mandatory education validated competency in understanding the content of the guidelines for all units and confirmed that the learning objectives were met.

A nursing grand rounds, fall case review, also introduced the Post-Fall Decision Guidelines. At our nursing grand rounds, we presented data from the Patient Safety Department on the severity of our falls and provided staff with the definition of how our falls are ranked. The National Patient Safety Goals or Never Events related to falls were discussed, and the need for accuracy when using the Morse Fall Scale, to assess patients' risk of falls, was stressed. Our standard interventions and exit plan to reduce the risk of falls were reviewed (Figure 2). The Post-Fall Decision Guidelines were presented, and physical therapy and occupational therapy representatives demonstrated safe transfer techniques. We also presented a video skit of 2 falls, one of a 33-year-old woman who developed a subdural hematoma after her fall and another of a 77-year-old man who fractured his clavicle and hip from 2 different falls. The video was an overdramatization of falls; the audience thoroughly enjoyed the video while reinforcing our fall initiatives.

Following the PDCA cycle, after approximately a year and additional evaluation of the patients who had fallen, complimentary tools were identified that could provide meaningful improvements and are being considered as a "fall bundle." The "fall bundle" includes a Post-Fall order set, revised fall policy, and postfall debriefing huddle. The PDCA cycle will be repeated once fully implemented.

The Post-Fall order set was developed and made available to physicians who assess patients after they have fallen. The order set was developed by a multidisciplinary team and includes standards of care for patients with fall-related injuries to facilitate consistency of

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FALL INTERVENTION ORDER SET FOR PATIENTS WHO HAVE FALLEN		Order entered Place ✓	RN audit Initial
ORDERS IN THIS SECTION WILL BE INITIATED UNLESS A LINE IS DRAWN THROUGH CONTENT			
1.	Notify charge nurse, complete assessment and refer to Post Fall Decision Guidelines		
2.	Perform baseline assessment and then a focused assessment every 2 hours times 4 hours		
ADDITIONAL ORDERS – CHECK THE BOXES TO ACTIVATE THE ORDERS			
3.	**Post fall Monitoring (documentation of fall evaluation to occur in progress note)** **Is patient on anticoagulants/antiplatelets? ☐ Yes ☐ No** **If yes,** **Hold the following anticoagulant medications**_____ ☐ **Sequential compression device to legs for VTE prophylaxis**		
4.	☐ **For a potential head injury** **Diagnostic** ☐ STAT non contrast head CT scan; call results ☐ Nurse to accompany patient ☐ PT/INR ☐ PTT ☐ CBC ☐ Other_____ **Nursing Interventions** • Neurological checks and vital signs every 15 minutes times 4, every 30 minutes times 4, every 2 hours times 8 **Call physician for** • Nausea and vomiting • Diastolic Blood pressure greater than _____or less than _____. • Systolic blood pressure greater than_____or less than_____ • Heart rate greater than _____or less than _____ • Change in neurological status		
5.	☐ **For Potential fracture** ☐ STAT X – ray of _____ ☐ PT/INR ☐ PTT ☐ CBC ☐ Other_____ **Nursing Interventions** • Vital signs and circulation checks every hour for 4 hours, then routine • Immobilize suspected limb **Call physician for**: • Blood pressure greater than _____ or less than_____ • Heart rate greater than _____ or less than _____ • Changes in circulation checks		
6.	☐ **Other Injury** Diagnostic_____ Nursing Interventions_____ Assessment_____ Call physician for _____		
7.			
8.			
9.			

Practitioner's Signature Date: Time:	Practitioner's Name or Dictation #	RN audit completed. Date/time/first initial/Last name/title

PATIENT IDENTIFICATION LABEL

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White copy – chart Yellow copy – Pharmacy\
Need mnemonic 9/09

Figure 3. Fall intervention order set for patients who have fallen. CT indicates computed tomography; PT, prothrombin time; INR, international normalized ratio; CBC, complete blood cell count; and PTT, partial thromboplastin time. Reprinted with permission. Copyright 2009 TriHealth, Inc.

care (Figure 3). The postfall debriefing huddle is a gathering of the charge nurse and nursing staff involved in the care of the patient who has fallen to assess factors that may have contributed to the fall. The huddle takes place after the patient has been stabilized but before the end of the shift for staff involved. The purpose of the huddle is to address those

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Date of fall	Time of fall	Pt ID No.	MFS scoring at time of fall	Pt's age	Admitting diagnosis	Secondary diagnosis	Level of alertness at time of fall	Fall interventions in place at time of fall	Pt alone at time of fall	Family with pt at time of fall	In bed, chair, ambulating	Was there a pt need at time of fall (eg, bathroom, reaching for something)	Time since pt last checked	Roll belt in use

Pt, patient

Figure 4. Postfall huddle form. Reprinted with permission. Copyright 2009 TriHealth, Inc.

factors that contributed to the fall and that can be controllable to prevent another fall. Postfall huddles increase staff awareness of a patient's fall risk. The findings from the huddle are documented on a separate form (Figure 4).

SUMMARY

While much literature addresses fall prevention, there is limited information on the care of a patient after a fall has occurred. Noting this gap offered an opportunity for improvement. After identifying variations in care after patients had fallen, the goal was to clarify expectations by providing direction for notification, structured communication, targeted assessments that include prompts for identifying changes in condition, and documentation. Minimal training was required, and there was no cost involved. The Post-Fall Decision Guidelines serve as a proactive tool to assist nurses in caring for patients in a consistent manner. While not all falls can be prevented, the guidelines ensure that all patients who fall will receive the appropriate care and assessment.

REFERENCES

1. Centers for Disease Control and Prevention. Falls among older adults; summary of research findings. http://cdc.gov/ncipc/pub-res/toolkit/SummaryOfFalls.htm. Published September 7, 2006. Accessed July 29, 2008.NOT_FOUND
2. Thomas KE, Stevens JA, Sarmiento K, Wald MM. Fall-related traumatic brain injury deaths and hospitalizations among older adults—United States 2005. *J Safety Res.* 2008;39(3):269-272. http://www.cdc.gov/ncipc/tbi/article_for_web.pdf. Accessed June 30, 2008.18571567
3. The Joint Commission. Sentinel Event Statistics. http://www.jointcommission.org/SentinelEvents/Statistics. Accessed February 12, 2009.
4. Nolan T, Resar R, Haraden C, Griffin FA. *Improving the Reliability of Health Care*. IHI Innovation Series white paper. Boston, MA: Institute for Healthcare Improvement; 2004. http://www.ihi.org/IHI/Results/WhitePapers/ImprovingtheReliabilityofHealthCare.htm. Accessed January 6, 2009.
5. Reason J. Combating omission errors through task analysis and good reminders. *Qual Saf Health Care.* 2002;11(1):40-44.NOT_FOUND;INVALID_JOURNAL
6. Grout J. *Mistake-Proofing the Design of Health Care Processes* (Prepared under an IPA with Berry College). Rockville, MD: Agency for Healthcare Research and Quality; 2007. AHRQ Publication No. 07-0020. http://www.ahrq.gov/qual/mistakeproof/. Accessed January 6, 2009.
7. Franko J, Kish KJ, O'Connell BG, Subramanian S, Yuschak JV. Advanced age and preinjury warfarin anticoagulation increase the risk of mortality after head trauma. *J Trauma.* 2006;61(1):107-110.16832256
8. Institute for Healthcare Improvement. SBAR technique for communication: a situational briefing model. General safety: tools. http://www.ihi.org/

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IHI/topics/PatientSafety/SafetyGeneral/Tools/SBARTechniqueforCommunication. Published 2008. Accessed February 12, 2009.
9. Joint Commission Resources. National Patient Safety Goals. http://www.jcrinc.com/National-Patient-Safety-Goals/. Accessed February 12, 2009.
10. Woodhall LJ, Vertacnik L, McLaughlin M. Implementation of the SBAR communication technique in a tertiary center. *J Emerg Nurs*. 2008;34(4):314-317. 18640411
11. Poe SS, Cvach MM, Gartrell DG, Radzik BR, Joy TL. An evidence-based approach to fall risk assessment, prevention, and management. *J Nurs Care Qual*. 2005;20(2):107-116.15841556
12. Kramer M, Schmalenberg C. Development and evaluation of essentials of magnetism tool. *J Nurs Adm*. 2004;34(6):365-378.15303055
13. Tague NR. *The Quality Toolbox*. 2nd ed. Milwaukee, WI: ASQ Quality Press; 2004:390-392. http://www.asq.org/learn-about-quality/project-planning-tools/overview/pdca-cycle.html. Accessed December 10, 2009.

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